Design of Spinodal Microstructures in High-Dimensional Composition Spaces via Analytic Derivatives of CALPHAD Model Predictions

Courtney Kunselman[a,*], Doğuhan Sarıtürk[a], Siya Zhu[a], Vahid Attari[a], Raymundo Arróyave[a,b,c]

[a]*Department of Materials Science and Engineering, Texas A&M University, College Station, 77843, TX, USA*
[b]*J. Mike Walker '66 Department of Mechanical Engineering, Texas A&M University, College Station, 77843, TX, USA*
[c]*Wm Michael Barnes '64 Department of Industrial and Systems Engineering, Texas A&M University, College Station, 77843, TX, USA*

Abstract

Keywords:

1. Introduction

Spinodal decomposition is a phase separation process characterized by barrierless uphill diffusion resulting in a bicontinuous microstructure morphology with periodic oscillations in composition. Since properties arise from structure, it is important to identify regions of conditions space which may be unstable to composition fluctuations and susceptible to spontaneous phase separation when designing alloys. In some cases, spinodal decomposition is undesirable and can lead to significant hardening and embrittlement of single-phase ductile alloys [1] or provide a a pathway for the formation of brittle, deleterious phases [2]. On the other hand, spinodal and spinodal-assisted microstructures have recently been leveraged to overcome the strength-ductility tradeoff [3–5], tailor magnetic properties [6, 7], and improve thermoelectric performance [8].

Regardless of whether the designer is looking to exploit or avoid spinodal decomposition, such a region must first be located within a design space that is often high-dimensional. Because exhaustive experimental characterization of these large design spaces is infeasible, computational modeling approaches which can predict microstructure stability are essential for the efficient identification of spinodal regions. Recent work to this end includes high-throughput *ab initio* calculations coupled with cluster expansion to construct CALPHAD-style Gibbs energy models [9] and a novel procedure not requiring any functional form of the free energy in which the spinodal decomposition process is modeled as a nucleation event infinitesimally close to the spinodal boundary [10]. While these approaches do not require any empirical data and were well-validated with experiments, they were only demonstrated on binary systems. Showing promise for spinodal design in high-dimensional composition spaces, Koneru, Kadirvel, and Wang recently developed a CALPHAD-based high-throughput microstructure stability predictive framework specifically for the multi-principal element alloy (MPEA) or high entropy alloy (HEA) design problem [11, 12]. This framework, now implemented in

*Corresponding author
Email address: cjkunselman18@tamu.edu (Courtney Kunselman)

Pandat [13], facilitates gauge-invariant prediction and visualization of summaries of Gibbs energy surface geometry such as eigenvalues and eigenvectors of the Gibbs energy Hessian, spinodal temperatures, and initial composition fluctuations.

In this work, we start with the framework of Koneru, Kadirvel, and Wang as a foundation and demonstrate a few key extensions. First, rather than using numerical derivative approximations which can become computationally expensive as the number of components gets large, we employ the recently formalized Jansson derivative method to calculate analytic Gibbs energy Hessians in a robust and efficient manner [14]. Second, instead of relying on access to commercial databases which may not contain models for all alloy systems of interest, we use the recently developed PhaseForge package [15] to automatically construct multicomponent CALPHAD models from machine learning-based interatomic potentials (MLIP). Third, we include an estimate for the elastic energy contributions due to coherency strains which penalize phase separation and effectively shrink the miscibility gap and spinodal region. Fourth, instead of using a collection of constant core component (a type of pseudoternary) diagrams to visualize results, we employ affine projections [16] to summarize high-dimensional data on a single two-dimensional plot.

Lastly, the entire workflow is packaged in an open-source tool available through the MaterialsFramework interface [17]. Through this framework, modelers have whitebox access to the CALPHAD model, and as demonstrated in [18], they can further refine parameters to any available computational or experimental data through ESPEI [19]. The remainder of this work will dive into the theory and methodology of determining microstructure stability from the Gibbs energy description of a phase and then demonstrate our framework to investigate the stability of the BCC phase in the Hf-Nb-Ti-V system.

2. Theory

2.1. Developing Gibbs Energy Models From Universal Potentials

The stability analysis described in the following section requires a molar Gibbs energy description of the solid solution phase of interest as a function of composition and temperature. Classical CALPHAD databases are resource-intensive to develop and may not exist for all alloy systems of interest. To overcome this limitation, we employ the PhaseForge package [15], which automates the construction of CALPHAD-compatible Gibbs energy models from MLIPs as described in [20, 21]. This approach replaces computationally expensive density functional theory (DFT) calculations with MLIP evaluations that are several orders of magnitude faster while maintaining thermodynamic accuracy.

The approach is built on the use of special quasirandom structures (SQS) [22] to represent disordered solid solutions in a periodic cell. An SQS is a periodic supercell whose short-range order parameters match those of a perfectly random alloy up to a pre-determined neighbor range, enabling the disordered state to be represented without requiring large supercells.

Following the compound energy formalism (CEF) as implemented in the Alloy Theoretic Automated Toolkit (ATAT) [23], the molar Gibbs energy of solution phase β with site fractions $\boldsymbol{y}$ at temperature T is given by

$$G_m^{\beta}(\boldsymbol{y}, T) = \left(G_{\text{calc,nc}}^{\beta}(\boldsymbol{y}, T) - \sum_i x_i \, G_{\text{calc,nc}}^{\alpha_i}(T) \right) + \sum_i x_i \, G_{\text{SGTE}}^{\alpha_i}(T) - T S_{\text{id}}(\boldsymbol{y}), \tag{1}$$

where $G_{\text{calc,nc}}^{\beta}$ is the Gibbs energy of phase β computed from MLIP total energies without any configurational entropy contribution, $G_{\text{calc,nc}}^{\alpha_i}$ is the corresponding quantity for pre component i in its reference crystal structure α_i, $G_{\text{SGTE}}^{\alpha_i}$ is the unary Gibbs energy of component i taken from the Scientific Group Thermodata Europe (SGTE) database [24], x_i is the mole fraction component i, and $S_{\text{id}}(\boldsymbol{y})$ is the ideal configurational entropy. For a single-sublattice solution phase, the ideal configurational entropy reduces to

$$S_{\text{id}}(\boldsymbol{y}) = -R \sum_i x_i \ln x_i, \tag{2}$$

where R is the ideal gas constant.

For each SQS supercell, the quantity $G_{\text{calc,nc}}^{\beta}$ at 0 K is taken directly from the MLIP-relaxed potential energy per atom. Structural relaxations are performed through the MaterialsFramework calculator interface [17] with a force convergence criterion of $f_{\max} < 0.001$ eV/Å. For the BCC_A2 phase in the Hf−Nb−Ti−V system, the compositional dependence of the mixing energy is captured sufficiently by static formation energies, and thermal contributions are accounted for through the SGTE unary data.

Once formation energies have been computed for SQS supercells spanning the composition space of each phase, the composition-dependent mixing contribution embedded in $G_{\text{calc,nc}}^{\beta}$ is parameterized using a Redlich-Kister polynomial expansion. For a binary A−B subsystem, the excess Gibbs energy is expressed as

$$G_{m,\text{xs}}^{\beta} = x_A x_B \sum_{\nu=0}^{k} L_{A,B}^{(\nu)} \, (x_A - x_B)^{\nu}, \tag{3}$$

where $L_{A,B}^{(\nu)}$ are the ν-th order binary interaction parameters determined by least-squares fitting to the MLIP-computed formation energies. Temperature-independent interaction parameters are assumed throughout. Higher-order ternary and quaternary interactions are incorporated analogously and are extracted by fitting to SQS energies at compositions spanning the multicomponent space.

The fitted interaction parameters are assembled into a thermodynamic database (TDB) file compatible with PyCalphad [25]. The resulting TDB for the BCC_A2 phase in the Hf−Nb−Ti−V system serves as the thermodynamic input for the stability analysis described below.

2.2. Determining Stability From the Gibbs Energy Hessian

Gibbs first introduced the concept of a thermodynamic stability limit, or spinodal, by considering composition fluctuations leading to continuous changes in fluid phases [26]. He determined that the limit of stability for such a

phase relative to these fluctuations occurs where the derivatives of the chemical potential of each component with respect to the density of its corresponding component are equal to zero. That is, a system stable to all composition fluctuations experiences positive changes in all of its chemical potentials when the content of the corresponding component is increased. As Cahn explains in [27], for binary system $A - B$ where B is the dependent component, this limit of stability simplifies to the well-known expression

$$\frac{d^2 G_m^\alpha}{dx_A^2} = 0, \tag{4}$$

where G_m^α is the molar Gibbs energy for phase α and x_A is the composition of component A.

Following Gibbs and Cahn, the pioneering work of de Fontaine rigorously derived the conditions for stability for solid phases in multicomponent systems by including coherency stress contributions to the free energy in both continuum [28] and microscopic [29] contexts. In recent works, Kadirvel *et al.* provide an elegant condensed proof of these conditions using the continuum formulation [12], and Morral and Chen apply de Fontaine's findings to comprehensibly extend Gibbs' work to the solid phase, multicomponent system context. We summarize their combined insight below.

Consider solid solution phase α, an n-component phase with composition $\boldsymbol{x}$ for which we define the n-th component as the dependent composition variable. Let $\boldsymbol{\Delta}$ be a local arbitrary composition fluctuation in phase α defined for all independent composition variables. To maintain mass balance, a local composition fluctuation of $-\boldsymbol{\Delta}$ must also occur for an equivalent amount of the system. The molar Gibbs energy change of the portion of phase α affected by this composition fluctuation is given by

$$\Delta G_m^\alpha = \frac{1}{2} G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta}) + \frac{1}{2} G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta}) - G_m^\alpha(\boldsymbol{x}). \tag{5}$$

Using a second order Taylor series expanded around $\boldsymbol{x}$, we can approximate $G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta})$ as

$$G_m^\alpha(\boldsymbol{x} + \boldsymbol{\Delta}) \approx G_m^\alpha(\boldsymbol{x}) + \sum_{i=1}^{n-1} \frac{\partial G_m^\alpha}{\partial x_i} \Delta_i + \frac{1}{2} \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} \frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j} \Delta_i \Delta_j \tag{6}$$

and $G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta})$ as

$$G_m^\alpha(\boldsymbol{x} - \boldsymbol{\Delta}) \approx G_m^\alpha(\boldsymbol{x}) - \sum_{i=1}^{n-1} \frac{\partial G_m^\alpha}{\partial x_i} \Delta_i + \frac{1}{2} \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} \frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j} \Delta_i \Delta_j \tag{7}$$

where all derivatives are evaluated at $\boldsymbol{x}$.

Substituting Equations (6) and (7) into Equation (5) gives

$$\Delta G_m^\alpha = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} \frac{\partial^2 G_m^\alpha}{\partial x_i \partial x_j} \Delta_i \Delta_j = \boldsymbol{\Delta}^T H \boldsymbol{\Delta} \tag{8}$$

where H is the Hessian of G_m^α with respect to the independent composition variables. Assuming that all entries of the Hessian are continuous throughout the composition domain, we can claim that H is a symmetric matrix with real

eigenvalues $\lambda_1, \ldots, \lambda_{n-1}$ and a corresponding set of eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n-1}$ which form an orthonormal basis in the space defined by the independent composition variables. Thus, $\boldsymbol{\Delta}$ can be written as a linear combination of this basis:

$$\boldsymbol{\Delta} = \sum_{i=1}^{n-1} (\boldsymbol{\Delta}^T \boldsymbol{v}_i) \boldsymbol{v}_i. \tag{9}$$

Substituting Equation (9) into Equation (8) and leveraging the relationship between eigenvalues and eigenvectors ($H\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$) gives

$$\begin{aligned} \Delta G_m^{\alpha} &= \left(\sum_{i=1}^{n-1} (\boldsymbol{\Delta}^T \boldsymbol{v}_i) \boldsymbol{v}_i \right)^T H \left(\sum_{i=1}^{n-1} (\boldsymbol{\Delta}^T \boldsymbol{v}_i) \boldsymbol{v}_i \right) \\ &= \left(\sum_{i=1}^{n-1} (\boldsymbol{\Delta}^T \boldsymbol{v}_i) \boldsymbol{v}_i \right)^T \left(\sum_{i=1}^{n-1} (\boldsymbol{\Delta}^T \boldsymbol{v}_i) \lambda_i \boldsymbol{v}_i \right) = \sum_{i=1}^{n-1} \lambda_i (\boldsymbol{\Delta}^T \boldsymbol{v}_i)^2. \end{aligned} \tag{10}$$

Thus, as Equation (10) illustrates, such a composition fluctuation can only lower the Gibbs energy if at least one of the eigenvalues is negative, and a zero-valued eigenvalue represents the limit of stability, or the spinodal. If only one eigenvalue, say λ_1, is negative, the system is conditionally unstable to composition fluctuations in the direction of $\boldsymbol{v}_1$, or equivalently, with large values for the projection $\boldsymbol{\Delta}^T \boldsymbol{v}_1$. More negative eigenvalues lead to more directions for $\boldsymbol{\Delta}$ to be unstable, and $n - 1$ negative eigenvalues imply that the solid solution is unstable to any $\boldsymbol{\Delta}$.

As Morral and Chen [30] and Kadirvel *et al.* [12] both report, these eigenvalues and eigenvectors are not gauge invariant. That is, $\lambda_1, \ldots, \lambda_{n-1}$ and $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n-1}$ depend on the choice of the dependent component. This is due to a combination of the coordinate axes being geometrically non-orthogonal and the change in orientation of these axes with the variation of the dependent component.

To address this issue, Kadirvel *et al.* [12] followed the approach of Singh *et al.* [31] which involves a change of basis operation transforming H from Gibbs simplex space to Cartesian space, performing the eigenanalysis in Cartesian space, and lastly transforming the resulting eigenvectors back to Gibbs space (the eigenvalues calculated in Cartesian space remain invariant under this final transformation). Singh *et al.* [31] provide an iterative procedure for constructing such an N-dimensional linear transformation, and Kadirvel *et al.* [12] explicitly provide this matrix for systems with ten components or fewer. In this work, we adopt this approach in order to calculate and visually display the distribution of gauge-invariant eigenvalues.

Lastly, for arbitrarily-complex phases modeled with internal degrees of freedom (or equivalently, for any phase with a Gibbs energy function that is multivalued in composition space), calculating the Gibbs energy Hessian requires differentiating the outputs of single-phase energy minimization calculations. These equilibrium calculations prescribe the composition of the phase (not the system) and determine the local minimum solution (which may be metastable) under this composition constraint. For phase α in ternary system A-B-C, it is well known that the first derivative of G_m^{α} with respect to x_A can be written as a simple linear combination of chemical potentials

$$\frac{\partial G_m^\alpha}{\partial x_A} = \mu_A - \mu_C \tag{11}$$

where C is the dependent component. Thus, second order derivatives can be expressed as

$$\frac{\partial^2 G_m^\alpha}{\partial x_B \partial x_A} = \frac{\partial \mu_A}{\partial x_B} - \frac{\partial \mu_C}{\partial x_B}, \tag{12}$$

but calculation of Equation (12) is less trivial. In this work, we employ the recently formalized Jansson derivative method [14], an adjoint-like technique for calculating analytic derivatives of thermodynamic properties at equilibrium with respect to CALPHAD model equilibrium calculations, to efficiently calculate the derivatives of the chemical potentials in Equation (12) rather than relying on numerical approximations. All Gibbs energy Hessians are computed using PyCalphad [25] (development version commit a8de09f).

Using the procedure described above, we can employ Jansson derivatives to calculate the chemical Hessian and corresponding chemical spinodal from a CALPHAD Gibbs energy description of a solution phase. However, in many solid solutions, composition fluctuations can lead to significant elastic energy contributions due to coherency strains. These coherency strains penalize composition fluctuations and when their elastic energy contributions are added to the chemical Gibbs energy, the spinodal region shrinks. Siya: We have to clarify that we assume the coherence is "biaxial" and the system is "infinite isotropic".As Cahn shows in [27], for an infinite binary isotropic solid with an arbitrary composition fluctuation Siya: I think the equation given by Cahn works only for infinitesimal coherent composition fluctuations instead of arbitrary or finite-amplitude composition fluctuations. He dropped the higher order terms in equation(2) and wrote "for small (c-c0)" above equation(7) in his paper. , when coherency strains are considered the limit of stability given in Equation (4) becomes

$$\frac{d^2 G_m^\alpha}{dx_A^2} + \frac{2\eta_A^2 E V_m}{1-\nu} = 0 \tag{13}$$

where E is the modulus of elasticity, V_m is the molar volume, ν is the Poisson ratio, and η_A is the linear strain per unit difference in component A:

$$\eta_A = \frac{1}{a}\frac{\partial a}{\partial x_A} \tag{14}$$

where a is the lattice parameter. Extending to the multicomponent context [32], entries of the coherent Hessian H^{coh} are given by

$$H_{ab}^{coh} = \frac{\partial^2 G_m^\alpha}{\partial x_B \partial x_A} + \frac{2\eta_A \eta_B E V_m}{1-\nu} \tag{15}$$

where $\frac{\partial^2 G_m^\alpha}{\partial x_B \partial x_A}$ is the chemical contribution as calculated in Equation (12).

In this work, we employ the bond-based lattice parameter prediction model Siya: why don't we use the lattice parameters from the relaxed SQS directly? Courtney: because we wanted an analytic model for the lattice parameter

that we could easily differentiate to calculate eta. developed by Tandoc *et al.* [33] to calculate $\boldsymbol{\eta}$ and V_m. Under this model, the average lattice parameter $\bar{a}$ for a solid solution BCC phase is given by

$$\bar{a} = \frac{2\bar{d}_{111}}{\sqrt{3}} \tag{16}$$

where $\bar{d}_{111}$ is the average bond length along the $< 111 >$ close-packed direction. The average bond length is calculated through

$$\bar{d}_{111} = \sum_i \sum_j x_i x_j d^{ij}_{111} \tag{17}$$

where d^{ij}_{111} is the first nearest neighbor (FNN) bond length for an $i-j$ pair along the $< 111 >$ close-packed direction, following the framework of Tandoc *et al.* [33]. Same-species bond lengths d^{ii}_{111} are extracted from MLIP-relaxed pure BCC cells via $d^{ii}_{111} = a^{\mathrm{BCC}}_i \sqrt{3}/2$, while cross-species lengths d^{ij}_{111} $(i \neq j)$ are obtained from MLIP-relaxed B2 (CsCl-type) binary ordered cells via $d^{ij}_{111} = a^{\mathrm{B2}}_{ij} \sqrt{3}/2$. All relaxations for bond length extraction are performed using the `BondLatticeParameter` tool within MaterialsFramework [17] with the GRACE-2L-SMAX-OMAT-large universal MLIP. Using this model for BCC structures, the molar volume is then given by

$$V_m = \frac{\bar{a}^3 N_A}{2} \tag{18}$$

where N_A is Avogadro's number. To calculate derivatives of the lattice parameter for η, a dependent component must be assigned which must remain consistent throughout the calculation of the coherent Hessian. Let k be the index of the dependent component. Then

$$x_k = 1 - \sum_{i \neq k} x_i, \tag{19}$$

and

$$\bar{d}_{111} = \sum_{i\neq k}\sum_{j\neq k} x_i x_j d^{ij}_{111} + \sum_{i\neq k} x_i (1 - \sum_{j\neq k} x_j) d^{ik}_{111} + \sum_{i\neq k} (1 - \sum_{j\neq k} x_j) x_i d^{ki}_{111} + (1 - \sum_{j\neq k} x_j)^2 d^{kk}_{111}. \tag{20}$$

Differentiating Equation (20) with respect to independent composition variable x_1 gives

$$\begin{aligned} \frac{\partial \bar{d}_{111}}{\partial x_1} &= \sum_{i\neq k} (x_i d^{i1}_{111} + x_i d^{1i}_{111}) - \sum_{i\neq 1,k} (x_i d^{ik}_{111} + x_i d^{ki}_{111}) + (1 - 2x_1 - \sum_{i\neq 1,k} x_i) d^{1k}_{111} \\ &\quad + (1 - 2x_1 - \sum_{i\neq 1,k} x_i) d^{k1}_{111} - 2(1 - \sum_{i\neq k} x_i) d^{kk}_{111} \\ &= 2\left(\sum_{i\neq k} x_i d^{i1}_{111} - \sum_{i\neq 1,k} x_i d^{ik}_{111} + (x_k - x_1) d^{1k}_{111} - x_k d^{kk}_{111} \right). \end{aligned} \tag{21}$$

Differentiating Equation (16) and plugging in the result from Equation (21), we have

$$\frac{\partial \bar{a}}{\partial x_1} = \frac{4}{\sqrt{3}} \left(\sum_{i \neq k} x_i d^{i1}_{111} - \sum_{i \neq 1,k} x_i d^{ik}_{111} + (x_k - x_1) d^{1k}_{111} - x_k d^{kk}_{111} \right). \tag{22}$$

Lastly, the elastic constants C_{11}, C_{12}, and C_{44} were computed for each composition of interest using the `CubicElasticConstantsAnalyzer` within the MaterialsFramework package [17]. A $4 \times 4 \times 4$ BCC SQS supercell was first relaxed using the GRACE-2L-SMAX-OMAT-large universal MLIP with unconstrained cell shape to a force convergence of $f_{\max} < 0.01\,\mathrm{eV/Å}$. Three finite strain modes were then applied to the relaxed cell: a uniform volumetric distortion to extract the bulk modulus B, an orthorhombic distortion to isolate $C' = (C_{11} - C_{12})/2$, and a monoclinic shear to determine C_{44}. Polycrystalline-averaged elastic properties were then obtained via the Voigt-Reuss-Hill (VRH) scheme, yielding the macroscopically isotropic Young's modulus E and Poisson's ratio ν. The biaxial modulus $M = E/(1-\nu)$ appearing in the coherent Heissian above was derived directly from these VRH averages. Siya: Are the Young's moduli and Poisson's ratio assumed isotropic? The Cahn's model requires the isotropy. How does materialsframework deal with that? For bcc structure the isotropy is described as $A = \frac{2C_{44}}{C_{11}-C_{12}}$ and is isotropic when $A = 1$. For Nb, given by "Elastic Constants of Niobium from 4.2 to 300K", K. Carroll, J. Appl. Phys., 1965., at 300K, the C_{11} is $24.56 \times 10^{11} dyn/cm^2$, the C_{12} is $13.87 \times 10^{11} dyn/cm^2$, and the C_{44} is $2.930 \times 10^{11} dyn/cm^2$, $A = 0.548$. The Young's moduli and Poisson ratio calculated based on the experimental data are:

Table 1: Direction-dependent Young's modulus and Poisson's ratio of single-crystal Nb at 300 K, calculated from the experimental elastic constants $C_{11} = 245.6$ GPa, $C_{12} = 138.7$ GPa, and $C_{44} = 29.30$ GPa.

Loading direction	Transverse direction	E (GPa)	ν
[100]	[010]	145.5	0.361
[110]	[1$\bar{1}$0]	93.2	0.591
[110]	[001]	93.2	0.231
[111]	[1$\bar{1}$0]	83.2	0.420

How we decide the E and ν in Equation 12?

2.3. *Verifying Microstructure Instability with Phase-Field Modeling*

While the eigenanalysis of the Gibbs energy Hessian described in Section 2.2 identifies *whether* a composition is spinodally unstable and in *which* compositional direction decomposition is favored, it does not reveal the temporal evolution of the resulting microstructure. To bridge this gap, we couple the CALPHAD-derived thermodynamics directly to a phase-field simulation based on the Cahn-Hilliard equation [27, 34].

2.3.1. Free Energy Functional and Governing Equations

For an n-component alloy described by $n-1$ independent composition fields $x_i(\boldsymbol{r}, t)$, $i = 1, \ldots, n-1$, with the n-th component serving as the reference (dependent variable), the total free energy functional is

$$\mathcal{F}[\boldsymbol{x}] = \int_\Omega \left[G_m^\alpha(\boldsymbol{x}) + \sum_{i=1}^{n-1} \frac{\kappa_i}{2} \left| \nabla x_i \right|^2 \right] dV, \tag{23}$$

where $G_m^\alpha(\boldsymbol{x})$ is the molar Gibbs energy of phase α (the CALPHAD-derived quantity described in Section 2.1), and κ_i (J m^2 mol^{-1}) is the gradient energy coefficient for field i that penalizes sharp compositional interfaces. The time evolution of each composition field is governed by the Cahn-Hilliard equation:

$$\frac{\partial x_i}{\partial t} = \nabla \cdot \sum_{j=1}^{n-1} M_{ij}(\boldsymbol{x}) \, \nabla \mu_j^{\mathrm{tot}}, \qquad i = 1, \ldots, n-1, \tag{24}$$

where $\boldsymbol{M}(\boldsymbol{x})$ is the composition-dependent mobility tensor (m^5 J^{-1} s^{-1}) and μ_j^{tot} is the variational chemical potential:

$$\mu_j^{\mathrm{tot}} = \frac{\delta \mathcal{F}}{\delta x_j} = \underbrace{\frac{\partial G_m^\alpha}{\partial x_j}}_{\mu_j - \mu_n} - \kappa_j \nabla^2 x_j. \tag{25}$$

The bulk driving force $\mu_j - \mu_n = \partial G_m^\alpha / \partial x_j$ is precisely the relative chemical potential already established in Equation (11). To prevent diffusion from diverging as any component approaches a pure-element limit, we employ a degenerate mobility,

$$M_{ij}(\boldsymbol{x}) = \mathring{M}_{ij} \prod_{k=1}^{n} x_k, \tag{26}$$

where $\mathring{M}_{ij}$ is the composition-independent (bare) mobility matrix. The product over all n mole fractions vanishes whenever any $x_k \to 0$, enforcing physical boundary behaviour.

Equation (24) is solved numerically on a uniform $N_g \times N_g$ grid (domain size L_x) using a pseudo-spectral semi-implicit scheme [35]. Diagonal mobility terms are treated implicitly (providing unconditional damping of high-wavenumber modes) while off-diagonal cross-terms are treated explicitly:

$$\hat{x}_i^{n+1} = \frac{\hat{x}_i^n + \Delta t \, k^2 \sum_j M_{ij}^{\mathrm{eff}} \, \hat{\mu}_{j,\mathrm{bulk}}^n - \Delta t \, k^4 \sum_{j \neq i} M_{ij}^{\mathrm{eff}} \, \kappa_j \, \hat{x}_j^n}{1 + \Delta t \, k^4 \, M_{ii}^{\mathrm{eff}} \, \kappa_i}, \tag{27}$$

where $\hat{(\cdot)}$ denotes the discrete Fourier transform, k^2 is the squared wavenumber, and $M_{ij}^{\mathrm{eff}} = \mathring{M}_{ij} \langle \prod_k x_k \rangle$ is the spatially-averaged effective mobility. The stable time step is estimated from the Hessian of the free energy at the mean composition $\boldsymbol{x}_0$:

$$\Delta t_{\mathrm{stable}} = \frac{\alpha_{\mathrm{s}} \, \Delta x^2}{\max_i(\mathring{M}_{ii}) \, |\lambda_{\max}(H)|}, \tag{28}$$

where $\alpha_{\mathrm{s}} < 1$ is a safety factor, $\Delta x = L_x / N_g$, and $|\lambda_{\max}(H)|$ is the largest absolute eigenvalue of the Gibbs energy Hessian $H_{ij} = \partial^2 G_m^\alpha / \partial x_i \partial x_j$ evaluated in a neighbourhood of $\boldsymbol{x}_0$.

2.3.2. Eyre Convex-Splitting Stabilisation

The semi-implicit scheme of Equation (27) treats the gradient energy term $\kappa_i k^4$ implicitly, which unconditionally damps high-wavenumber oscillations arising from interfacial stiffness. However, within the spinodal region the diagonal Hessian elements $H_{ii} = \partial^2 G_m^\alpha / \partial x_i^2$ are *negative*, and the corresponding bulk contribution $M_{ii} H_{ii} k^2$ enters the numerator with a destabilising sign. For large $|\Delta t H_{ii}|$ this explicit term can cause unbounded growth and numerical blowup, regardless of the spatial resolution.

Convex-splitting decomposition.. Following Eyre [**?**], we decompose the molar Gibbs energy G_m^α into a *contracting* (convex) part G^+ and an *expanding* (concave) part G^-,

$$G_m^\alpha(\boldsymbol{x}) = G^+(\boldsymbol{x}) - G^-(\boldsymbol{x}), \qquad G^\pm \text{ convex}, \tag{29}$$

and treat G^+ implicitly and G^- explicitly in time. For a quadratic approximation to the bulk free energy at the mean composition $\boldsymbol{x}_0$, a sufficient convex-splitting for component i is

$$G_i^+ = \tfrac{A_i}{2}\, x_i^2, \qquad G_i^- = \tfrac{A_i}{2}\, x_i^2 - G_m^\alpha(\boldsymbol{x}), \tag{30}$$

where the *stabilisation constant* A_i must satisfy

$$A_i \;\geq\; \max\!\big(0,\, -H_{ii}(\boldsymbol{x}_0)\big) \tag{31}$$

to ensure that $G_i^+ - \frac{A_i}{2} x_i^2$ remains convex [**? ?**]. With this splitting, the update rule for field i in Fourier space becomes

$$\hat{x}_i^{n+1} = \frac{\hat{x}_i^n + \Delta t\, k^2\, M_{ii}^{\text{eff}} (\hat{\mu}_{i,\text{bulk}}^n - A_i\, \hat{x}_i^n) + \Delta t\, k^2\, \hat{s}_{\text{cross}}^n}{1 + \Delta t\, k^2\, M_{ii}^{\text{eff}}\, A_i + \Delta t\, k^4\, M_{ii}^{\text{eff}}\, \kappa_i}, \tag{32}$$

where $\hat{s}_{\text{cross}}^n$ collects all off-diagonal mobility cross-terms treated explicitly. The denominator is strictly positive for any $\Delta t > 0$, $A_i \geq 0$, $\kappa_i > 0$, making the scheme *unconditionally stable*: the solution cannot grow without bound regardless of the chosen time step.

Role of the stabilisation constant.. The choice of A_i controls the balance between stability and accuracy. To see this, note that the *effective* bulk driving force seen by field i in the numerator of Equation (32) is

$$H_{ii}^{\text{eff}} = H_{ii}(\boldsymbol{x}_0) + A_i. \tag{33}$$

Three regimes arise:

1. $A_i = |H_{ii}|$ **(minimum required).** $H_{ii}^{\text{eff}} = 0$ the linear spinodal driving force is exactly neutralised in the explicit part, and decomposition is driven purely by the *nonlinear* deviation of $\mu_{i,\text{bulk}}$ from its linearisation about $\boldsymbol{x}_0$. This is the theoretically optimal choice: the scheme is unconditionally stable and introduces no artificial damping of the spinodal growth rate.

2. $A_i > |H_{ii}|$ **(over-stabilised).** $H_{ii}^{\text{eff}} > 0$ the explicit update treats component i as if it were thermodynamically *stable*, actively opposing spinodal growth at every time step. The scheme remains unconditionally stable and will not blow up for any Δt, but the linear spinodal growth rate is artificially suppressed. Decomposition still eventually occurs, driven by the nonlinear chemical potential, but at a rate slower than the true physical dynamics.
3. $A_i < |H_{ii}|$ **(under-stabilised).** $H_{ii}^{\text{eff}} < 0$ the explicit term retains a net destabilising contribution. The scheme may blow up for large Δt and is only conditionally stable, recovering the classical CFL restriction.

Practical determination of A_i.. The condition Equation (31) is a lower bound, not a prescription. In practice, selecting A_i requires care because the diagonal Hessian $H_{ii}(\boldsymbol{x})$ is *not constant*: it depends on composition, temperature, and the alloy system. A value of A_i that is sufficient at the nominal composition $\boldsymbol{x}_0$ may become inadequate as the field evolves away from $\boldsymbol{x}_0$ during decomposition. Several strategies are possible.

1. **Point evaluation at $\boldsymbol{x}_0$.** The simplest approach sets $A_i = \max(0, -H_{ii}(\boldsymbol{x}_0))$. This is exact at the initial composition but may be insufficient once spinodal decomposition drives the local composition toward the equilibrium tie-line endpoints, where $|H_{ii}|$ can be substantially larger.
2. **Neighborhood scan (recommended).** Evaluate $H_{ii}(\boldsymbol{x})$ on a coarse grid spanning a window $[\boldsymbol{x}_0 \pm \delta]$ around the nominal composition, and take $A_i = \max_{\boldsymbol{x}\in\text{scan}} \max(0, -H_{ii}(\boldsymbol{x}))$. This anticipates the composition excursions during decomposition without evaluating the full spinodal region. In the present implementation, a window $\delta = 0.30$ (mole fraction) is used by default, sampled on a 20^d grid where d is the number of independent composition variables.
3. **Full spinodal scan.** Evaluate H_{ii} everywhere the diagonal element is negative (i.e. inside the spinodal boundary) and set A_i to the global minimum of H_{ii}. This guarantees unconditional stability for all reachable compositions but may over-stabilise the early dynamics.
4. **Adaptive A_i.** Recompute $A_i(t) = \max(0, -H_{ii}(\bar{\boldsymbol{x}}(t)))$ periodically from the current spatial mean $\bar{\boldsymbol{x}}(t)$ as the simulation progresses. This is theoretically optimal: A_i tracks the evolving composition and remains as close to $|H_{ii}|$ as possible at every stage, neither over-stabilising the early spinodal nor under-stabilising the late coarsening.

The sensitivity of A_i to composition can be significant. As a concrete example, for the binary $Nb_{50}V_{50}$ system at 500 °C, the diagonal Hessian element $H_{\text{Nb}}(\boldsymbol{x})$ takes the value $-7239\,\text{J mol}^{-1}$ at the nominal composition $x_{\text{Nb}} = 0.50$, but reaches $-19\,840\,\text{J mol}^{-1}$ near $x_{\text{Nb}} = 0.75$ an increase of nearly 2.7×. Relying solely on the point evaluation at $\boldsymbol{x}_0$ therefore underestimates the required stabilisation constant by that factor, causing the scheme to lose unconditional stability as the field composition evolves.

This sensitivity is also alloy- and temperature-dependent. In the ternary $Nb_{50}Hf_5V_{45}$ system, H_{Nb} varies from $0\,\text{J mol}^{-1}$ (neutrally stable) near 750 °C to $-5547\,\text{J mol}^{-1}$ at 600 °C and more negative values at lower temperatures, while H_{Hf} remains positive at all temperatures studied here. Consequently, the required A_i differs component by component, alloy by alloy, and temperature by temperature; there is no universal constant that applies across systems.

Per-component stabilisation.. A key feature of the present implementation is that A_i is set *independently for each component*. Elements with $H_{ii} > 0$ everywhere in the scan window (e.g. Hf in Nb-Hf-V at 600 °C, with $H_{\mathrm{Hf}} \approx 87\,720\,\mathrm{J\,mol^{-1}}$) receive $A_i = 0$ and contribute no extra stabilisation. Spinodally unstable elements (e.g. Nb, with $H_{\mathrm{Nb}} = -5547\,\mathrm{J\,mol^{-1}}$ at 600 °C) receive the minimum A_i needed to satisfy Equation (31) over the composition scan. This per-component assignment avoids introducing unnecessary artificial damping on stable species, which would slow their diffusional response and distort the coupled composition dynamics.

Interpolant boundary effects.. A practical issue distinct from numerical stability arises when composition field values overshoot the boundary of the pre-computed thermodynamic table. The surrogate interpolant is constructed on a finite composition grid; if x_i^n at any spatial point lies outside that grid, the interpolated chemical potential is undefined (typically returning `NaN`). Because the update Equation (32) is evaluated in Fourier space, a single `NaN` propagates instantaneously to every grid point, halting the simulation. This is *not* a stability failure of the Eyre scheme it is a thermodynamic evaluation error and it cannot be resolved by adjusting Δt or A_i.

The correct remedy is to evaluate the chemical potential on a *clamped copy* of the composition field, $\boldsymbol{x}^{\mathrm{eval}} = \mathrm{clamp}(\boldsymbol{x}^n, \epsilon, 1-\epsilon)$, rescaled to satisfy the simplex constraint, while updating the PDE field $\boldsymbol{x}^{n+1}$ from the unclamped $\boldsymbol{x}^n$. The stored composition field is never modified, so mass conservation is preserved exactly. The clamped evaluation is only invoked for points that would otherwise lie outside the interpolant domain; in the interior of the spinodal region these corrections are negligible.

Consequence for the time-step safety factor.. Because the Eyre scheme is unconditionally stable, it does not blow up even when the time step Δt exceeds the classical stability limit $\Delta t_{\mathrm{stable}}$ of Equation (28). The safety factor α_{s} in Equation (28) therefore controls *accuracy* rather than *stability*: larger α_{s} gives a coarser temporal resolution of the early-stage spinodal amplification and coarsening dynamics, but will not produce a divergent solution. In practice, values $\alpha_{\mathrm{s}} \leq 1$ give accurate reproduction of the linear spinodal growth rate, while values $\alpha_{\mathrm{s}} \gg 1$ can be used to fast-forward to late-stage microstructural states at the cost of temporal fidelity in the early regime.

2.3.3. CALPHAD-Coupled Surrogate for Thermodynamic Driving Forces

Direct evaluation of pycalphad at every spatial grid point and every time step is computationally prohibitive (each equilibrium call requires an iterative minimization). Instead, all thermodynamic quantities are pre-computed on a dense composition grid spanning the relevant simplex, stored in an HDF5 file, and replaced by smooth surrogate functions that are evaluated in $O(\log N)$ time during simulation.

At each grid point $\boldsymbol{x}^{(k)}$, three quantities are evaluated once using pycalphad's `IsolatedPhase` framework, which constrains every calculation to the single-phase $\mathrm{BCC_{A2}}$ free energy surface: the molar Gibbs energy $G_m^{\alpha,(k)}$ ($\mathrm{J\,mol^{-1}}$); the absolute chemical potential $\mu_i^{(k)}$ of each element i ($\mathrm{J\,mol^{-1}}$); and the Jansson second derivative $\Lambda_{ij}^{(k)} = \partial\mu_i/\partial x_j$ [14] ($\mathrm{J\,mol^{-1}}$), which is the analytic derivative of the chemical potential with respect to composition at single-phase

equilibrium. All three datasets are stored in a structured HDF5 file and loaded once at the start of a simulation run; no further thermodynamic calculations are performed during time integration.

The surrogate is constructed using composition-appropriate interpolation backends: a cubic spline for binary systems ($n = 2$), a C^1-smooth Clough-Tocher triangular interpolant [36] for ternary systems ($n = 3$), and a two-stage thin-plate-spline RBF-to-regular-grid scheme [37] for quaternary and higher-order systems ($n \geq 4$). All three methods share the same surrogate class and are used identically for G_m^α, μ_i, and Λ_{ij}.

2.3.4. Derivative Methods for the Bulk Chemical Potential

Computing the bulk driving force $\mu_j - \mu_n = \partial G_m^\alpha / \partial x_j$ and the Hessian H_{ij} from the surrogate can be done in three ways, selectable by the user. The choice matters because the Hessian controls both the spinodal wavelength $\lambda^* = 4\pi\sqrt{\kappa_i/|H_{ij}|}$ and the stable time step (Eq. 28), so derivative errors propagate directly into microstructural length and time scales.

Method I Finite differences on G_m^α (`fd`).. The chemical potential and Hessian are obtained numerically from the G_m^α surrogate by central finite differences:

$$\left(\mu_j - \mu_n\right)^{\mathrm{fd}} = \frac{G_m^\alpha(\boldsymbol{x} + \varepsilon \boldsymbol{e}_j) - G_m^\alpha(\boldsymbol{x} - \varepsilon \boldsymbol{e}_j)}{2\varepsilon} + O(\varepsilon^2), \tag{34}$$

and similarly for H_{ij} using the standard second-order stencil, where $\boldsymbol{e}_j$ is the unit vector in the j-th composition direction and $\varepsilon = 10^{-4}$. This approach requires only the G_m^α table; no chemical potential data need be stored. However, truncation error compounds with the smoothing bias of the surrogate, and errors in H_{ij} are amplified near the simplex boundaries where the ideal-entropy contribution $-RT\sum_i x_i \ln x_i$ makes G_m^α highly curved.

Method II Direct Jansson interpolation (`jansson`).. Under the simplex constraint $x_n = 1 - \sum_{i=1}^{n-1} x_i$, the Gibbs-Duhem equation gives [38]:

$$\mu_j - \mu_n = \left.\frac{\partial G_m^\alpha}{\partial x_j}\right|_{\text{simplex}}, \tag{35}$$

and the Hessian element is (from Equation (12)):

$$H_{ij} = \left.\frac{\partial^2 G_m^\alpha}{\partial x_i \, \partial x_j}\right|_{\text{simplex}} = \Lambda_{ij} - \Lambda_{nj}, \tag{36}$$

where $\Lambda_{ij} = \partial \mu_i / \partial x_j$ is the Jansson second derivative [14] pre-computed and stored in the HDF5 table. Separate surrogate functions are constructed for each $\mu_j - \mu_n$ (from Eq. 35) and each H_{ij} (from Eq. 36) using identical backends. This method is free of finite-difference truncation error and satisfies the Gibbs-Duhem relation exactly at every queried composition. For an n-component system, it requires $(n-1)$ additional surrogates for the chemical potentials and $(n-1)^2$ for the Hessian components a total of $(n-1)n$ interpolants beyond the G_m^α surface all constructed once at simulation startup.

Method III Hybrid: Jansson μ, finite-difference H (`hybrid`).. The bulk driving force $\mu_j - \mu_n$ is obtained from the Jansson surrogates (Eq. 35), while the Hessian H_{ij} is obtained by finite differences on G_m^α (Eq. 34). This combination is appropriate when the Jansson derivative dataset (`dMU`) is unavailable or when pycalphad's numerical Hessian data are unreliable near a phase boundary, while still providing thermodynamically consistent chemical potentials for the time-evolution equation.

2.3.5. Validation of Derivative Methods

Figure 1 compares the three modes against raw CALPHAD data for the Nb-V BCC_{A2} binary system at $T = 500\,°C$ as a function of the vanadium mole fraction x_V. Panel (a) confirms that the molar Gibbs energy G_m^α is identical across all modes, since it is always evaluated from the same G_m^α surrogate. Panel (b) shows the relative chemical potential $\mu_{Nb} - \mu_V$: the `fd` mode deviates visibly from the CALPHAD reference (RMSE = $302\,J\,mol^{-1}$) while `jansson` and `hybrid` reproduce the stored values exactly (RMSE = $0\,J\,mol^{-1}$). Panel (c) shows the most striking difference: the Hessian $\partial^2 G_m^\alpha / \partial x_{Nb}^2$ is displayed with the y-axis clipped to $\pm 150\,kJ\,mol^{-1}$ (the ideal-entropy divergence at $x_V \to 0$ or 1 is annotated); the `fd` mode yields an RMSE of $442\,kJ\,mol^{-1}$ relative to the CALPHAD reference, entirely due to truncation error amplification near the simplex boundaries, whereas `jansson` matches exactly. Since the spinodal decomposition wavelength scales as $\lambda^* \propto |H_{ij}|^{-1/2}$, Hessian errors of this magnitude would translate into erroneous microstructural length-scale predictions; the `jansson` method eliminates this source of error.

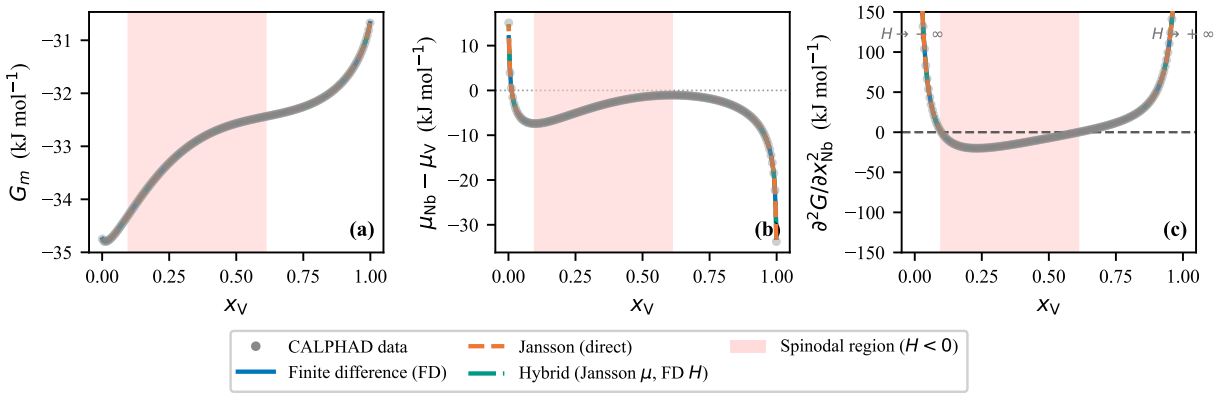


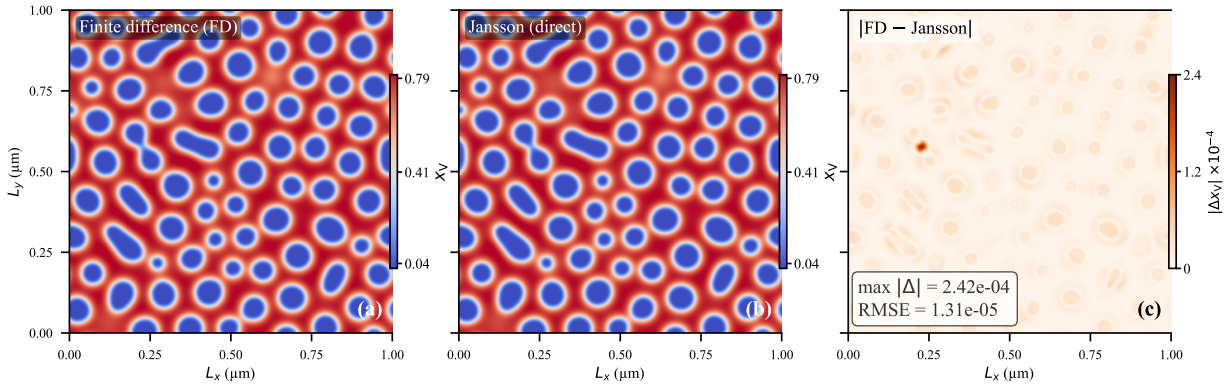
Figure 1: Comparison of the three derivative methods for the Nb-V BCC_{A2} system at $T = 500\,°C$. Grey markers: raw CALPHAD data (60% transparent to reveal underlying curves). (a) Molar Gibbs energy G_m^α: all three modes are identical (shared surrogate). (b) Relative chemical potential $\mu_{Nb} - \mu_V$: `fd` deviates by RMSE = $302\,J\,mol^{-1}$; `jansson` and `hybrid` are exact. (c) Hessian $\partial^2 G_m^\alpha / \partial x_{Nb}^2$ (clipped to $\pm 150\,kJ\,mol^{-1}$; divergence at the pure-element limits is annotated): `fd` RMSE = $442\,kJ\,mol^{-1}$; `jansson` is exact. The shaded pink band marks the spinodal region ($H < 0$) identified from the `jansson` Hessian.

To confirm that point-wise derivative agreement translates into equivalent field evolution, we ran two identical Cahn-Hilliard simulations — one each for the `fd` and `jansson` modes — on the Nb-V BCC_{A2} binary at $T = 500\,°C$ using a 256×256 periodic domain of side $L = 1$ ţm, $\kappa = 1\times10^{-12}\,J\,m^{-1}$, $M = 5\times10^{-21}\,m^3\,s\,kg^{-1}$, and 80 000 time steps of $\Delta t = 0.01\,s$ ($t = 800\,s \approx 26\,\tau$, where $\tau \approx 31\,s$ is the spinodal growth time constant at the dominant wavenumber

Table 2: Summary of derivative methods for bulk chemical potentials and Hessian in the CALPHAD-coupled Cahn-Hilliard model. RMSE values are for the Nb-V BCC_{A2} binary at $T = 500\,°C$ ($n = 200$ composition grid points). Bold: exact agreement with CALPHAD data.

Mode	$\mu_j - \mu_n$ source	H_{ij} source	HDF5 data required	$RMSE_\mu$ (J mol^{-1})	$RMSE_H$ (J mol^{-1})
`fd`	FD on G_m^α (Eq. 34)	FD on G_m^α	`/GM`	302	442 219
`jansson`	Eq. 35	Eq. 36	`/GM`, `/MU`, `/dMU`	**0**	**0**
`hybrid`	Eq. 35	FD on G_m^α	`/GM`, `/MU`	**0**	442 219

$k^* = \sqrt{|H_{\min}|/4\kappa}$). Both runs were seeded identically; the only variable was the derivative method used to evaluate the chemical-potential driving force $\delta F/\delta c$. Figure 2 presents the final x_V composition fields and the absolute per-pixel difference map. After 800 s of evolution the field is fully phase-separated, with well-coarsened Nb-rich ($x_V \approx 0.02$) and V-rich ($x_V \approx 0.82$) domains of characteristic size ~ 200 nm. The two micrographs are visually identical; `fd` shows a small residual relative to `jansson` (RMSE $= 1.3 \times 10^{-5}$, max $|\Delta x_V| = 2.4 \times 10^{-4}$) concentrated at domain interfaces where the composition gradient is steepest and FD truncation error is most amplified. This tiny residual does not affect the phase morphology or domain statistics, confirming that both derivative modes are effectively equivalent for the binary system. Differences may grow for higher-order systems where scattered-data interpolants are used; those cases are addressed in **??**.



Method	min x_V	max x_V	RMSE vs Jansson	Simulation time (s)
FD	0.0190	0.8188	1.31e-05	279
Jansson	0.0190	0.8188	—	199

Figure 2: Microstructure comparison of the `fd` and `jansson` derivative modes for the Nb-V BCC_{A2} binary at $T = 500\,°C$ after 80 000 time steps ($t = 800$ s, 256×256 grid, $L = 1$ μm, initial composition $c_0 = x_V = 0.50$). (a) Finite-difference and (b) Jansson direct composition fields x_V; fully coarsened V-rich (red) and Nb-rich (blue) spinodal domains are visible. (c) Absolute difference map $|x_V^{fd} - x_V^{jansson}|$, showing that discrepancies are confined to interfaces (RMSE $= 1.31 \times 10^{-5}$, max $= 2.42 \times 10^{-4}$). The table summarises composition statistics and simulation wall times for each derivative method (FD: 279 s; Jansson: 199 s).

3. Investigating Spinodal Decomposition in the Hf-Nb-Ti-V System

In this section, we employ our stability map framework to investigate spinodal decomposition in the BCC phase of the Hf−Nb−Ti−V system. In a recent experimental study, single-phase BCC was observed for $Hf_{0.10}Nb_{0.40}Ti_{0.40}V_{0.10}$, and spinodal decomposition was reported for an alloy at the equiatomic composition [3]. Siya: Do we consider the HCP phase? HCP is the stable phase for Hf and Ti under room temperature. Courtney: So far, we have only been investigating the stability of BCC structures. We could make a note that we recognize that pure Hf and Ti are HCP at room temp, but for this demonstration, we are just looking at the BCC phase in isolation. High lattice strains around 5 % were also reported for one of the phases resulting from spinodal decomposition, indicating that accounting for coherent strain is extremely important when modeling spinodal decomposition in this system.

Note on how the CALPHAD model was generated, if necessary

Through the procedure outlined in the previous section, we calculated eigenvalues for the chemical and coherent Gibbs energy Hessians at various temperatures with a sampling density in composition space of 5 at.%. Figure 3 displays affine projections of these results at 600 K, 800 K and 1000 K. Any missing values represent convergence failures, which are most common when the composition of at least one component is dilute. A limitation of affine projections is that multiple compositions can map to the same position. To ensure that marker differences are visible, smaller markers are plotted on top of larger markers (e.g. a smaller green circle with a blue ring around it indicates that compositions mapped to that position exhibit both zero and one negative eigenvalue). At 600 K, spinodal shrinkage due to the addition of coherent strain contributions is most pronounced at compositions high in Hf or V. The region of two negative eigenvalues from the chemical Hessian also vanishes when coherency is taken into account. At 800 K, the chemical Hessian predicts spinodal decomposition at almost all compositions except for those rich in Ti and Nb whereas the coherent Hessian predicts BCC solid solutions rich in Hf and V to be more stable to composition fluctuations. Lastly, at 1000 K, the previous pattern continues along with eigenvalues of the coherent Hessian all remaining positive for BCC alloys rich in Ti and V.

The consistent pattern of coherent strain contributions significantly suppressing the spinodal in regions rich in Hf and low in Ti and Nb could be explained by examining the components of the first eigenvector of the chemical Hessian. Figure 4 displays the components of this vector for a BCC alloy with composition $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$ as a function of temperature from 600 K to 1000 K. Note that throughout this temperature range, first eigenvalues of the chemical Hessian are negative while all eigenvalues of the coherent Hessian are positive. Thus, the components of the eigenvector displayed in Figure 4 represents the direction of greatest composition instability for a BCC solid solution when coherent strain contributions are not considered. We see that such an initial composition fluctuation would result in regions rich in Hf and low in V and vice versa. However, Hf and V have the largest atomic radius mismatch of all four components, and such a composition fluctuation could cause large coherency strains. Consequently, when an elastic contribution describing coherent strain resulting from composition fluctuations is added to the Gibbs energy at this composition, the local curvature of the Gibbs energy is adjusted and the alloy becomes stable to the previously

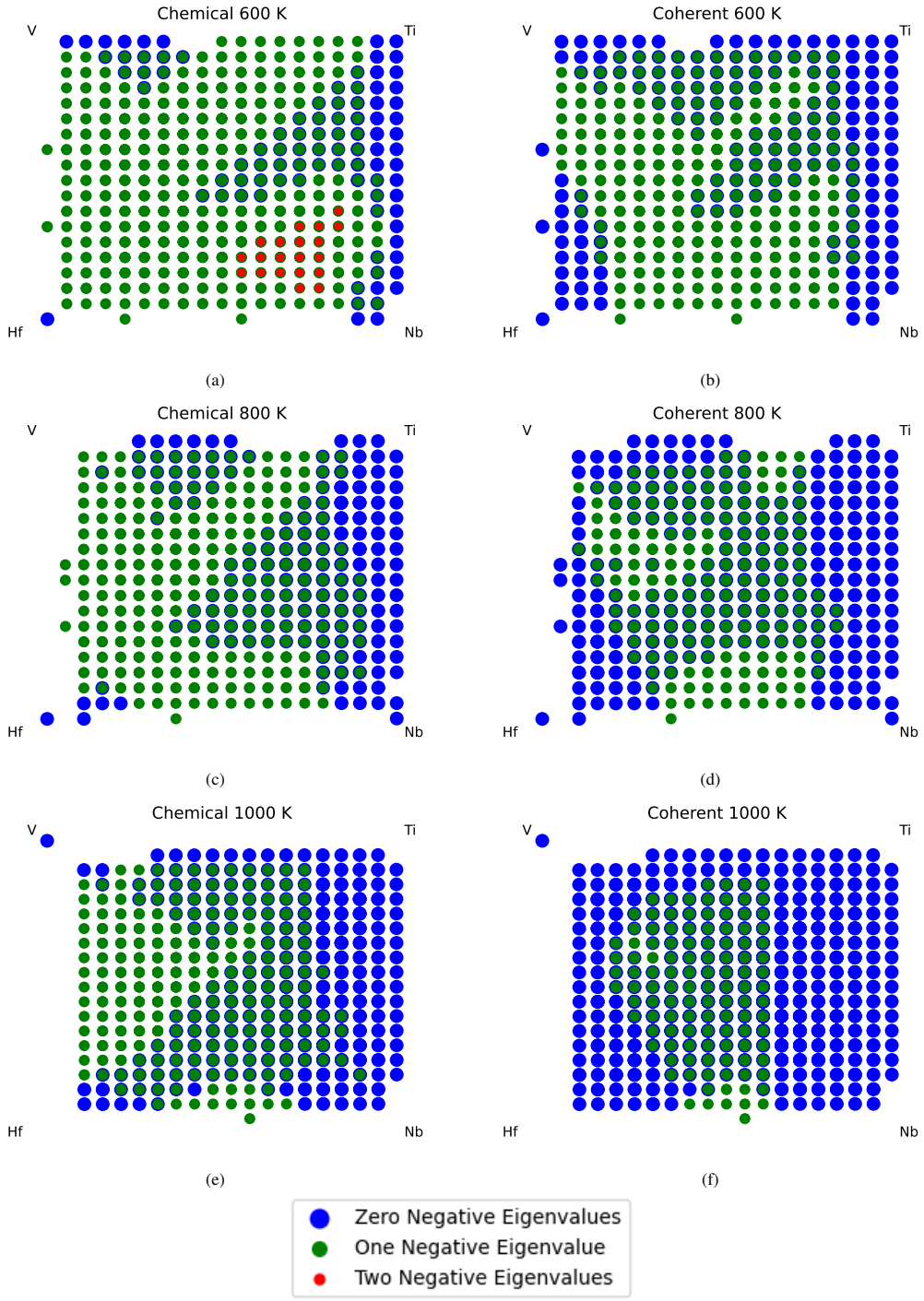


Figure 3: Affine projection stability maps over composition space displaying the number of negative eigenvalues for the chemical and coherent Hessian of the BCC phase in the Hf−Nb−Ti−V system at (a-b) 600 K, (c-d) 800 K, and (e-f) 1000 K.

unstable composition fluctuation.

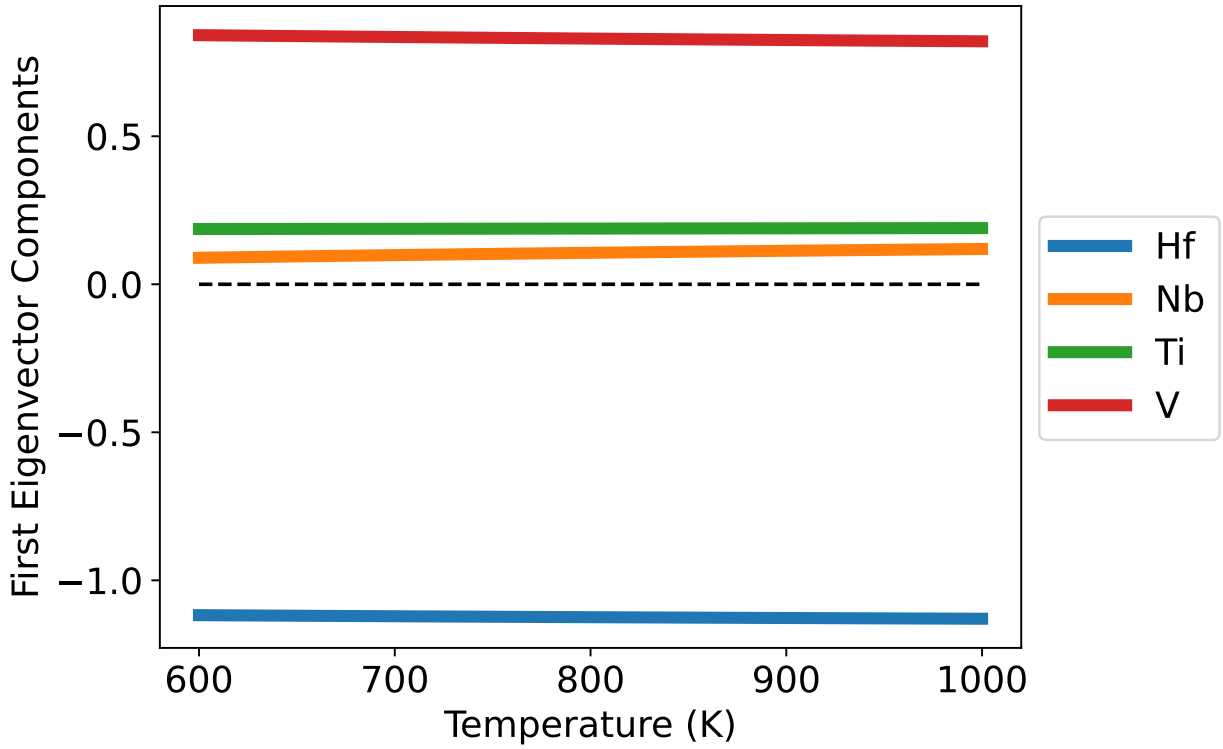


Figure 4: Components of the first gauge-invariant eigenvector of the chemical Hessian as a function of temperature for BCC alloy with composition $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$.

Eigenvalue investigations can also provide more insight. In Figure 5, we have the first eigenvalue for the chemical and coherent Hessian plotted as a function of temperature at compositions (a) $Hf_{0.10}Nb_{0.40}Ti_{0.40}V_{0.10}$, (b) $Hf_{0.25}Nb_{0.25}Ti_{0.25}V_{0.25}$, and (c) $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$. For the composition rich in Ti and Nb, coherent strain contributions have little effect on the first eigenvalue, and the spinodal temperatures are almost equivalent. As the composition of Hf increases, coherent strain has a greater effect on the spinodal temperature, leading to more stable solid solutions in Hf rich areas of composition space.

3.1. Thermodynamic Stiffness and Its Effect on Phase-Field Simulations

The stability maps and eigenvalue analysis above identify *which* compositions are spinodally unstable, but they also carry a practical consequence for phase-field simulations that is easy to overlook. The same Gibbs energy Hessian $\mathbf{H}$, whose negative eigenvalues signal spinodal instability, also controls the numerical stability of the time integrator. Specifically, in a semi-implicit spectral Cahn-Hilliard solver the maximum allowable time step is bounded by the *largest* Hessian eigenvalue:

$$\Delta t \;\leq\; \frac{C_s\,\Delta x^2}{M\,H_{\max}}, \tag{37}$$

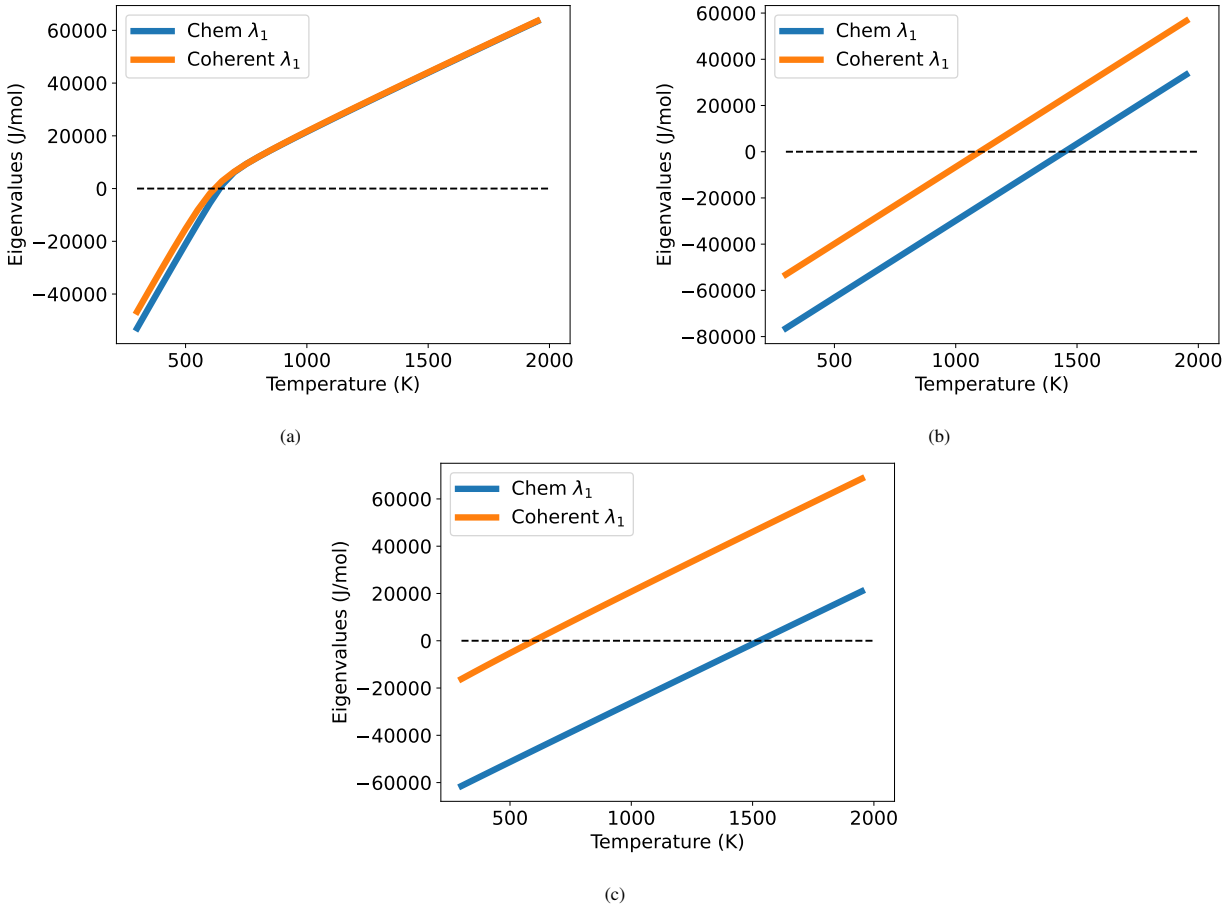


Figure 5: Gauge-invariant first eigenvalue as a function of temperature for the chemical and coherent Hessian of the BCC phase at compositions (a) $Hf_{0.10}Nb_{0.40}Ti_{0.40}V_{0.10}$, (b) $Hf_{0.25}Nb_{0.25}Ti_{0.25}V_{0.25}$, and (c) $Hf_{0.6}Nb_{0.1}Ti_{0.1}V_{0.2}$.

where C_s is a safety factor (here $C_s = 0.3$), Δx is the grid spacing, M is the atomic mobility, and $H_{\max}$ is the largest eigenvalue of $\mathbf{H}$ at the local composition. An element that is thermodynamically *stable* at the nominal alloy composition meaning its diagonal Hessian component $H_{ii} = \partial^2 G_m^\alpha / \partial x_i^2 > 0$ does *not* drive decomposition, but its large positive curvature increases $H_{\max}$ and therefore forces Δt down. We call such elements *numerical stiffeners*: they are spectators to the physical transformation yet they throttle the simulation's time-stepping and dramatically reduce the physical time accessible per unit of compute time.

It is important to emphasise that this is a *purely numerical* constraint. The thermodynamics is correct — the system is still globally spinodally unstable ($\lambda_{\min} < 0$) and will phase-separate if given sufficient physical time. The stiffeners simply make reaching that time computationally expensive.

Comparison across binary, ternary, and quaternary systems

To make this concrete, we compare four BCC_{A2} alloy compositions in the Hf-Nb-Ti-V system at $T = 500\,^\circ\mathrm{C}$, all simulated on a 256×256 grid with $L = 1\,\text{ţm}$, $\kappa = 1 \times 10^{-12}\,\mathrm{J\,m^{-1}}$, and $M = 5 \times 10^{-21}\,\mathrm{m^3\,s\,kg^{-1}}$. Table 3 summarises

the Gibbs energy Hessian at each nominal composition, the resulting time-step limit, and how much physical time is accessible within 50 000 integration steps.

Table 3: Gibbs energy Hessian analysis for four BCC_{A2} compositions at $T = 500\,°C$. Diagonal elements $H_i \equiv \partial^2 G_m^\alpha / \partial x_i^2$ are evaluated at the nominal composition; **negative values (bold)** indicate that the element is locally spinodally unstable and drives decomposition, while large positive values act as numerical stiffeners. $\lambda_{\min}$ is the minimum eigenvalue of the full Hessian matrix (negative = globally spinodal); $H_{\max}$ is the largest eigenvalue (the stiffness driver). Δt is the stability-limited time step (Eq. 37); τ is the spinodal growth timescale at the dominant wavenumber; t_{50k} is the physical time reached in 50 000 steps; and t_{50k}/τ is the number of growth timescales completed a direct measure of how much microstructural evolution has occurred. Simulation parameters: 256×256 grid, $L = 1$ ţm, $M = 5 \times 10^{-21}\,\mathrm{m^3\,s\,kg^{-1}}$.

System	H_{Nb}	H_{Hf}	H_{Ti}	$\lambda_{\min}$	$H_{\max}$	Δt (s)	τ (s)	t_{50k} (s)	t_{50k}/τ
			($\mathrm{J\,mol^{-1}}$)						
$Nb_{50}V_{50}$	**−9 263**	—	—	**−9 263**	9 263	1.0×10^{-2}	31	800	∼26
$Nb_{50}Hf_{10}V_{40}$	**−9 263**	**−68 756**	—	**−85 123**	85 123	1.3×10^{-2}	∼0	665	$\gg 10^3$
$Nb_{50}Hf_5Ti_5V_{40}$	**−9 207**	+254 911	+365 276	**−14 811**	376 302	1.32×10^{-3}	86	66	∼0.8
$Nb_{50}Hf_{2.5}Ti_{2.5}V_{45}$	**−8 217**	+410 350	+531 846	**−12 228**	543 681	1.32×10^{-3}	104	66	∼0.6

Figure 6 directly visualises the contrast in microstructural evolution between the binary, ternary, and first quaternary compositions.

Physical interpretation of the Hessian roles

Each diagonal Hessian element H_{ii} characterises the local free-energy curvature along a single-component fluctuation direction. A **negative** H_{ii} means the free energy is concave with respect to fluctuations in x_i: the system lowers its free energy by separating into regions richer and poorer in element i, which is the classical spinodal driving force. A **large positive** H_{ii} means the free energy is steeply convex: the system strongly resists composition fluctuations in x_i, and any such fluctuation incurs a large free-energy penalty.

In the Cahn-Hilliard PDE, both types of curvature enter on equal footing through the mobility-weighted Hessian $M\,\mathbf{H}$. The unstable modes ($H_{ii} < 0$) provide the driving force for decomposition; the stable modes ($H_{ii} > 0$) provide thermodynamic restoring forces. However, at the *numerical* level, the stable modes impose a stringent constraint on Δt through Eq. (37): the solver must take small enough steps that the steepest uphill curvature does not cause oscillatory or divergent behaviour. This is analogous to a stiff ODE where fast, stable modes force a small step size even if the slow, unstable mode of interest evolves on a much longer timescale.

The critical insight from Table 3 is that **reducing the Hf and Ti content does not relieve the stiffness**. Halving both additions from 5 to 2.5 at.% ($Nb_{50}Hf_{2.5}Ti_{2.5}V_{45}$) moves these elements further from the spinodal in the binary Hf-V and Ti-V sub-systems, which *increases* their positive curvature (H_{Hf}: $+255\,000 \rightarrow +410\,000\,\mathrm{J\,mol^{-1}}$;

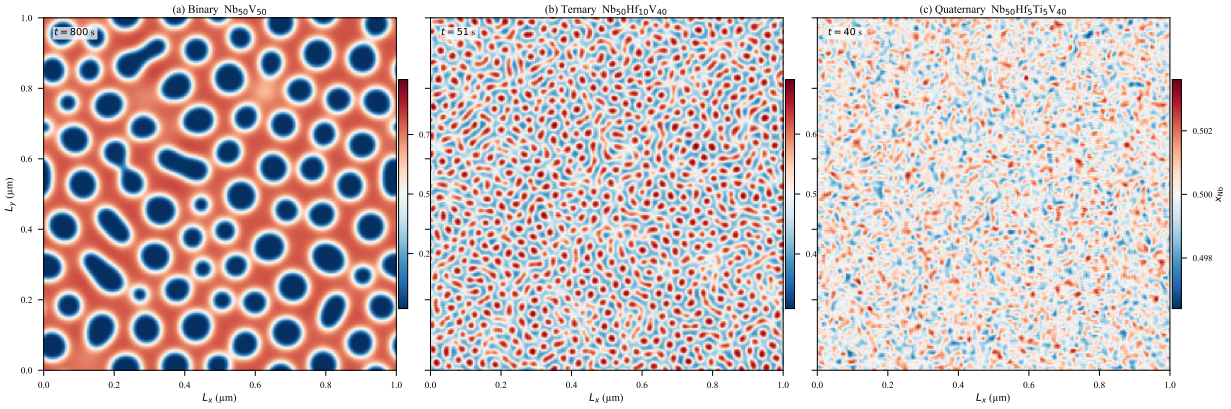


Figure 6: Composition fields from Cahn-Hilliard simulations (Jansson mode) of three BCC_{A2} alloys at $T = 500\,°C$ (256×256 grid, $L = 1$ ţm, $\kappa = 1 \times 10^{-12}\,J\,m^{-1}$, $M = 5 \times 10^{-21}\,m^3\,s\,kg^{-1}$). (a) Binary $Nb_{50}V_{50}$: x_V field after $t = 800$ s (80 000 steps, $\Delta t = 0.01$ s). Well-coarsened Nb-rich (blue) and V-rich (red) spinodal domains of characteristic size ~ 200 nm are visible. (b) Ternary $Nb_{50}Hf_{10}V_{40}$: x_{Nb} field after $t = 51$ s (30 000 steps, $\Delta t = 1.71 \times 10^{-3}$ s). Both Nb and Hf are spinodally unstable (H_{Nb}, $H_{Hf} < 0$), so there are no numerical stiffeners; the simulation completes $\gg 10^3\,\tau$ of evolution and clear fine-scale decomposition is observed (x_{Nb} spans [0.36, 0.69]). (c) Quaternary $Nb_{50}Hf_5Ti_5V_{40}$: x_{Nb} field after $t = 40$ s (30 000 steps, $\Delta t = 1.32 \times 10^{-3}$ s). Although globally spinodally unstable ($\lambda_{min} = -14\,811\,J\,mol^{-1}$), Hf and Ti act as numerical stiffeners ($H_{Hf} = +255\,kJ\,mol^{-1}$, $H_{Ti} = +365\,kJ\,mol^{-1}$), forcing a ten-fold reduction in Δt and limiting the simulation to only $\approx 0.8\,\tau$ far short of the $\gtrsim 10^2\,\tau$ required for visible domain formation. The composition field remains essentially uniform (x_{Nb} spans only [0.497, 0.503]).

H_{Ti}: $+365\,000 \rightarrow +532\,000\,J\,mol^{-1}$). The stable time step is essentially unchanged at $\Delta t = 1.32 \times 10^{-3}$ s, and the growth timescale τ actually *increases* from 86 to 104 s because the global minimum eigenvalue weakens slightly (λ_{min}: $-14\,811 \rightarrow -12\,228\,J\,mol^{-1}$). This is counter-intuitive: one might expect that diluting Hf and Ti toward zero recovers the fast binary Nb-V decomposition. However, it is the *absolute magnitude* of the positive curvature not the composition itself that controls stiffness, and this magnitude grows as the element moves deeper into the stable composition regime.

Implications for alloy and simulation design

This analysis reveals a general challenge in CALPHAD-coupled phase-field simulations of multi-component alloys. Whether an alloy will show spinodal decomposition in a phase-field simulation within a tractable number of steps depends on *two separate criteria* that must both be satisfied:

1. **Thermodynamic criterion:** the global minimum Hessian eigenvalue must be negative ($\lambda_{min} < 0$), confirming spinodal instability.
2. **Numerical accessibility criterion:** no individual element should have a large positive curvature $H_{ii} \gg 0$ that would suppress Δt below the level needed to reach $t \gtrsim 10^2\,\tau$ within a practical number of steps. Equivalently, all elements at the nominal composition should ideally reside *within* or close to the spinodal.

The stability maps of Figure 3 directly address criterion 1 by identifying spinodal compositions. They are equally useful for criterion 2: compositions deep inside the spinodal region for all components simultaneously will minimise

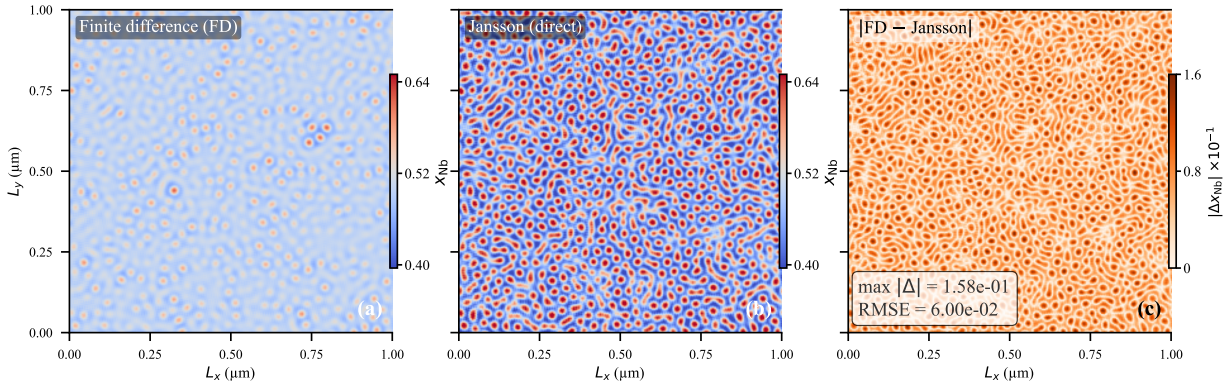
stiffness and maximise accessible physical time.

Temperature and composition dependence of spinodal character in Nb-Hf-V

The nature of the spinodal instability changes qualitatively with both temperature and Hf content, which has direct consequences for both alloy design and simulation strategy. To illustrate this, we computed the diagonal Hessian elements H_{Nb}, H_{Hf}, and $H_{\mathrm{V}} \equiv H_{\mathrm{Nb}} + H_{\mathrm{Hf}} + 2H_{\mathrm{Nb,Hf}}$ together with the minimum eigenvalue $\lambda_{\min}$ for the ternary $Nb_{50}Hf_5V_{45}$ alloy across temperatures from 500 °C to 1000 °C, and compared these against the binary $Nb_{50}V_{50}$ system. Table 4 summarises the resulting spinodal regimes.

The key contrast between the binary and ternary is that adding even 5 at.% Hf shifts the spinodal from a symmetric two-component instability to an asymmetric one-component instability (Nb only) across most of the accessible temperature range. At 700 °C, the ternary driving force ($|\lambda_{\min}| = 13\,272\,\mathrm{J\,mol^{-1}}$) is approximately six times weaker than at 500 °C ($|\lambda_{\min}| = 25\,166\,\mathrm{J\,mol^{-1}}$), making 700 °C the most numerically tractable temperature at which a physically meaningful Nb-driven spinodal decomposition still occurs in this system.

?? confirms this behaviour directly. Cahn-Hilliard simulations of $Nb_{50}Hf_5V_{45}$ were run at 500 °C and 700 °C under identical conditions (256 × 256 grid, $L = 1\,\text{ţm}$, $\kappa = 1 \times 10^{-12}\,\mathrm{J\,m^2\,mol^{-1}}$, $M_{\mathrm{Nb}} = 5 \times 10^{-21}\,\mathrm{m^5\,J^{-1}\,s^{-1}}$). At 500 °C, where both H_{Nb} and H_{V} are negative, clear Nb-rich and V-rich spinodal domains nucleate and coarsen in both composition fields. The Hf field closely tracks the Nb field, consistent with Hf partitioning passively into the Nb-rich phase rather than driving an independent instability. At 700 °C, where only $H_{\mathrm{Nb}} < 0$, the decomposition is weaker and the microstructure is less developed within the same number of steps, reflecting the reduced driving force and the asymmetric one-component character of the spinodal at this temperature.



Method	min x_{Nb}	max x_{Nb}	RMSE vs Jansson	Simulation time (s)
FD	0.4445	0.6161	6.00e-02	505
Jansson	0.3578	0.6912	—	364

Figure 7: Composition fields from Cahn-Hilliard simulations of the ternary $Nb_{50}Hf_5V_{45}$ BCC_{A2} alloy at 500 °C (top row) and 700 °C (bottom row) (256 × 256 grid, $L = 1\,\text{ţm}$, $\kappa = 1 \times 10^{-12}\,\mathrm{J\,m^2\,mol^{-1}}$, $M_{\mathrm{Nb}} = 5 \times 10^{-21}\,\mathrm{m^5\,J^{-1}\,s^{-1}}$, Jansson derivative mode). Columns show the x_{Nb}, x_{Hf}, and x_{V} composition fields at the final saved snapshot. At 500 °C, both Nb and V are individually unstable (H_{Nb}, $H_{\mathrm{V}} < 0$) and clear spinodal domains are visible; Hf partitions passively into the Nb-rich phase. At 700 °C, only Nb is unstable ($H_{\mathrm{Nb}} < 0$), yielding a weaker, asymmetric decomposition consistent with the reduced driving force ($|\lambda_{\min}|$ six times smaller than at 500 °C).

Table 4: Spinodal character and equilibrium two-phase compositions as a function of temperature for the binary $Nb_{50}V_{50}$ and ternary $Nb_{50}Hf_5V_{45}$ BCC_{A2} alloys. Signs of the diagonal Hessian elements indicate whether each element is individually unstable ($H_{ii} < 0$, driving force) or stable ($H_{ii} > 0$, spectator/stiffener). H_V is derived from the reduced Hessian via $H_V = H_{Nb} + H_{Hf} + 2H_{Nb,Hf}$.

Regime	System	Diagonal instabilities	Physical character	Equilibrium BCC phases[a]
$T < 750\,°C$	Binary $Nb_{50}V_{50}$	$H_{Nb} = H_V < 0$	Symmetric Nb↔V demixing; both components drive decomposition equally	BCC_1: $Nb_{99}V_1$ BCC_2: Nb_1V_{99}
$T > 700\,°C$	Ternary $Nb_{50}Hf_5V_{45}$	None ($H_{Nb}, H_{Hf}, H_V > 0$)	No diagonal instability; only weak off-diagonal Nb-Hf	Single-phase BCC: $Nb_{50}Hf_5V_{45}$

When stiff spectator elements cannot be avoided for example, when experimental alloy constraints fix the composition practical mitigation strategies include: (i) employing fully implicit or exponential-time-differencing (ETD) integrators, which are not bound by Eq. (37) and can take large steps even in the presence of stiff stable modes; (ii) operator-splitting, which handles the stiff stable modes analytically or implicitly while integrating the unstable spinodal modes explicitly; or (iii) rescaling time by recognising that the physical timescale of interest is τ, so reporting microstructural evolution in units of t/τ rather than absolute seconds makes meaningful comparisons across alloy systems regardless of stiffness.

4. Conclusions and Future Work

In this work, we demonstrated an open-source workflow for high-throughput microstructure stability analysis. Firstly, if a suitable CALPHAD model is not available to the modeler, it can be created through PhaseForge. The chemical Hessian is then determined analytically and efficiently via Jansson derivatives. Coherent strain contributions can be added through an analytically differentiable bond model and MLIP estimates for elastic constants. Lastly, following a clever change of basis, gauge-invariant eigenvalues and eigenvectors of the chemical and coherent spinodal can be investigated at specific compositions as functions of temperature or visualized at a given temperature over a high-dimensional composition space through affine projections. This workflow provides many valuable extensions to the foundational stability analysis framework produced by Koneru, Kadirvel, and Wang [11], to include 1) open-source access, 2) whitebox CALPHAD model knowledge, 3) improved efficiency and robustness of Gibbs energy derivatives, 4) modeling coherent strain contributions, and 5) visualization in large composition spaces via one two-dimensional plot.

However, there are still plenty of future directions to further extend this workflow. A model for homogeneous elastic Gibbs energy contributions could be incorporated in addition to contributions from coherent strain due to composition fluctuations. MLIPs could also be used to create bond models for other crystal structures, and coherent strain elastic contributions could be modeled for materials that do not behave isotropically. Lastly, implementation of second-order Jansson derivatives would provide analytic calculation of third-order derivatives of the Gibbs energy with respect to composition, enabling the analytic calculation of critical points.

Code Availiability

The code associated with this work is available at the following repository:

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